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Paper 30
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UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

DR. SQUATCH, LLC,
Petitioner,

v.

THE PROCTER & GAMBLE COMPANY,
Patent Owner.

IPR2024-01498
Patent 11,844,752 B2

Before JOHN G. NEW, SHERIDAN K. SNEDDEN, and SUSAN L. C. MITCHELL, *Administrative Patent Judges.*

SNEDDEN, *Administrative Patent Judge.*

JUDGMENT
Final Written Decision
Determining All Challenged Claims Unpatentable
35 U.S.C. § 318(a)

I. INTRODUCTION

A. Background and Summary

Dr. Squatch, LLC ("Petitioner") filed a Petition requesting *inter partes* review of claims 1–19 of U.S. Patent No. 11,844,752 B2 (Ex. 1001, "the '752 patent"). Petition ("Pet."), Paper 2. The Procter & Gamble Company ("Patent Owner") filed a Preliminary Response ("Prelim. Resp."). Paper 6.

We instituted trial on April 11, 2025. Paper 9 ("Inst. Dec."). During trial, Patent Owner filed a Patent Owner Response. Paper 13 ("PO Resp."). Petitioner filed a Reply (Paper 16 ("Pet. Reply")), and Patent Owner filed a Sur-reply (Paper 21 ("PO Sur-Reply")). We held an oral hearing on January 27, 2026, and a transcript of that hearing is of record. Paper 29 ("Tr.").

We have jurisdiction under 35 U.S.C. § 6(b). After considering the full record developed through trial, we determine that Petitioner has proved by a preponderance of the evidence that the challenged claims are unpatentable. *See* 35 U.S.C. § 316(e). Our reasoning is explained below, and we issue this Final Written Decision under 35 U.S.C. § 318(a).

B. Real Parties in Interest

Petitioner identifies Dr. Squatch, LLC, as its real party-in-interest. Pet. 1. Petitioner also advises the Board that Dr. Squatch, LLC is indirectly owned by Unilever PLC. Paper 18, 2.

Patent Owner identifies The Procter & Gamble Company as its real party-in-interest. Paper 4, 1.

C. Related Matters

The parties indicate that the ’752 patent is involved in *The Procter & Gamble Company v. Dr. Squatch LLC*, Case No. 2-24-cv-04711 (C.D. Cal.), filed November 29, 2022. Pet. 1; Paper 4, 1.

Petitioner has also filed petitions for *inter partes* review of the following patents: IPR2024-01104 against U.S. Patent No. 11,540,999; IPR2024-01105 against U.S. Patent No. 10,966,915; IPR2024-01173 against U.S. Patent No. 10,905,647; and IPR2024-01174 against U.S. Patent No. 11,497,706. Pet. 1; Paper 4, 1.

D. The ’752 patent (Ex. 1001)

The ’752 patent is titled “Deodorant Compositions.” Ex. 1001, code (54). The ’752 patent is directed to deodorant compositions and methods of making the compositions. *Id.* at 1:5–6. More specifically, the ’752 patent is directed to formulating, in response to consumer demands, an aluminum-free, mostly silicone-free deodorant stick, which is not too hard. *Id.* at 1:12–13, 1:21–23.

The ’752 patent explains that

> [c]onsumers also want a good glide, non-sticky, and non-greasy application. This is a challenge, because a mostly silicone-free formula will often use natural oils or natural oil-based triglycerides. But often natural oils, such as coconut oil, bring a large portion of wax with them, often making the resulting deodorant sticks too hard to meet a more preferable product application for both shaven and unshaven underarms. They are also often undesirably greasy due to the non-volatile nature.

Id. at 1:12–20. “As consumers seek more natural ingredients in their deodorants, one approach to formulation is to use emollients derived from natural oils.” *Id.* at 2:49–51. The specification describes that products aimed at emphasizing the use of natural ingredients "may have a significant amount of a liquid triglyceride." *Id.* at 2:56. To provide structure to these products, the '752 patent describes that it is known to include waxes and other such structurants. *Id.* at 2:58–60. However, the '752 patent explains that

> a currently marketed product, Comparative Formula 1 . . . , has 34.15% liquid triglyceride, along with a number of structurants, resulting in a very hard stick, scoring 63 on a needle penetration test under ASTM D-1321 . . . . So while Comparative Formula 1 uses consumer preferred natural ingredients, it does not necessarily provide a good consumer experience when used, given its hardness. Comparative Formula 2 contains even higher levels of structurants, resulting in a harder stick with an even lower hardness score. In comparison, Inventive Examples 1–5, while comprising consumer-preferred natural ingredients, have higher hardness scores, meaning they are softer products.

Id. at 2:62–3:7. According to the '752 patent, "a deodorant stick having at least about 25% of a liquid triglyceride, and that uses a primary structurant that has a melting point of at least about 50° C., . . . can result in a deodorant stick with a hardness from about 80 mm*10 to about 140 mm*10." *Id.* at 3:40–48. The '752 patent describes that the "deodorant stick is able to comprise consumer-perceived natural ingredients, while offering a pleasant consumer experience in terms of its hardness." *Id.* at 3:48–51.

The '752 patent describes that the deodorant compositions "can contain a structurant, an antimicrobial, a perfume, and additional chassis ingredient(s) . . . in the form of a solid stick." *Id.* at 9:11–14. The '752 patent defines a "structurant" as "any material known or otherwise effective in providing suspending, gelling, viscosifying, solidifying, and/or thickening properties to the composition or which otherwise provide structure to the final product form." *Id.* at 4:57–61. The '752 patent describes a deodorant stick comprising a magnesium hydroxide that is effective against underarm bacteria strains in raw material evaluations. *Id.* at 8:19–25, 8:34–36.

E. The Challenged Claims

Petitioner challenges claims 1–19. Representative independent claims 1 and 15 are reproduced below:

> 1. A deodorant stick comprising:
>
> a. at least one antimicrobial; and
>
> b. a structurant that is stearyl alcohol;
>
> said stick being free of an aluminum salt; and
>
> said stick having a hardness from about 80 mm*10 to about 140 mm*10, as measured by penetration with ASTM D-1321 needle;
>
> wherein the deodorant stick is anhydrous and is substantially free of silicones.
>
> 15. A deodorant stick comprising:
>
> a. at least about 25% by weight liquid triglyceride;
>
> b. a starch;
>
> c. stearyl alcohol;
>
> d. magnesium hydroxide;
>
> e. shea butter;
>
> f. coconut oil;
>
> g. triethyl citrate; and
>
> h. sunflower seed oil;
>
> said deodorant stick being free of an aluminum salt; and wherein said stick has a hardness from about 80 mm*10 to about 140 mm*10, as measured by penetration with ASTM D-1321 needle.

Ex. 1001, 15:26–34, 16:13–25.

F. Evidence

Petitioner relies upon information that includes the following.

Ex. 1003, Lesniak et al., US 11,433,018 B2, issued September 6, 2022 (“Lesniak”).

Ex. 1004, Phinney et al., US 9,314,412 B2, issued April 19, 2016 (“Phinney”).

Ex. 1005, John Henderson Lamb, M.D., *Sodium Bicarbonate: An Excellent Deodorant*, THE JOURNAL OF INVESTIGATIVE DERMATOLOGY 7(3) 131–33 (1946) (“Lamb”).

Ex. 1006, Bianchi et al., US 6,048,518, issued April 11, 2000 (“Bianchi ’518”).

Ex. 1007, Wayback Machine archives, captured May 26, 2016, November 23, 2016, and November 1, 2015 respectively for: https:/www.nativecos.com, https:/www.nativecos.com/product/deodorant/, and https:/www.nativecos.com (“Native”).

Ex. 1008, Bianchi, US 2007/0166254 A1, published July 19, 2007 (“Bianchi ’254”).

Ex. 1009, Wayback Machine archive, captured January 26, 2017, for http:/soapdelinews.com/2016/11/easy-homemade-deodorantrecipe.html (“Easy Homemade”).

Ex. 1044, Millet et al., US 2018/0168954 A1, published June 21, 2018 (“Millet”).

Ex. 1057, Anconi et al., U.S. Patent App. Pub. No. 2016/0089321, published March 31, 2016 (“Anconi”).

Ex. 1058, Stoia, et al., Selected Evidence-Based Health Benefits of Topically Applied Sunflower Oil, 10(1) APPLIED SCI. REPS. 45–49 (2015) (“Stoia”).

Ex. 1073, Sturgis et al., U.S. Patent App. Pub. No. 2019/0000747, published January 3, 2019 (“Sturgis”).

Petitioner also relies on the Declarations of Bozena B. Michniak-Kohn, Ph.D. (Exs. 1002, 1086) to support its contentions.

Patent Owner relies on the declaration of Robert Lochhead, Ph.D. (Ex. 2011) to support its contentions.

G. Asserted Grounds of Unpatentability

Petitioner asserts that claims 1–19 would have been unpatentable on the following grounds:

Ground	Claim(s) Challenged	35 U.S.C. §	Reference(s)/Basis
1	1–13	103	Lesniak, Bianchi '254
2	14	103	Lesniak, Bianchi '254, Phinney
3	15, 16, 18, 19	103	Lesniak, Bianchi '254, Phinney, Anconi
4	17	103	Lesniak, Bianchi '254, Phinney, Anconi, Easy Homemade
5	1, 4–14	103	Native, Bianchi '254
6	2–3	103	Native, Bianchi '254, Easy Homemade
7	15, 17–19	103	Native, Bianchi '254, Easy Homemade, Anconi, Stoia
8	16	103	Native, Bianchi '254, Easy Homemade, Anconi, Stoia, Millet
9	15–19	103	Squatch and Sturgis

H. Claim Construction

The challenged claims should be read in light of the Specification, as they would be interpreted by one of ordinary skill in the art. *In re Suitco Surface, Inc.*, 603 F.3d 1255, 1260 (Fed. Cir. 2010). Thus, we generally give claim terms their ordinary and customary meaning. *See In re Translogic Tech.*, Inc., 504 F.3d 1249, 1257 (Fed. Cir. 2007) ("The ordinary and customary meaning 'is the meaning that the term would have to a person of ordinary skill in the art in question.'" (internal quotation marks omitted)); *see also* 37 C.F.R. § 42.100(b) (stating that claims are construed in IPRs according to the same standard as used in federal court).

Petitioner does not argue that any claim terms require construction. *See generally* Pet.

Patent Owner asserts that the preamble of independent claims 1, 16, and 19 reciting "a deodorant stick" is limiting as it provides antecedent basis for the recitation of "said deodorant stick being free of an aluminum salt" for each independent claim. PO Resp. 7–10. Patent Owner also points to dependent claims 6–9 as relying on the preamble reciting "a deodorant stick" as antecedent basis for claim elements. *Id*. at 9. We agree. Petitioner does not argue to the contrary. *See*, *e.g*., Pet. 19 (arguing that Lesniak discloses a deodorant stick in the form of an antiperspirant stick).

We determine that no express construction of any claim term is necessary to determine whether to institute *inter partes* review. *Nidec Motor Corp. v. Zhongshan Broad Ocean Motor Co*., 868 F.3d 1013, 1017 (Fed. Cir. 2017) ("[W]e need only construe terms 'that are in controversy, and only to the extent necessary to resolve the controversy.'" (quoting *Vivid Techs., Inc. v. Am. Sci. & Eng'g, Inc*., 200 F.3d 795, 803 (Fed. Cir. 1999))). To the extent further discussion of the meaning any claim term is necessary to our decision, we provide that discussion below in our analysis of the asserted grounds of unpatentability.

I. Level of Ordinary Skill in the Art

Petitioner proposes that a person of ordinary skill in the art ("POSA" or "POSITA") at the time of the invention "would have (1) at least a B.S. or equivalent degree; and (2) at least two years of experience in cosmetics formulations." Pet. 17 (citing Ex. 1002 ¶¶ 26–28). Patent Owner does not dispute Petitioner's proposal about the POSA's qualifications at this stage of the proceeding. PO Resp. 10.

We adopt and apply Petitioner's proposed definition of a POSA, which is consistent with the level of skill reflected in the asserted prior art. *See e.g.*, Exs. 1003, 1004.

II. ANALYSIS

A. Introduction

"In an *inter partes* [review], the petitioner has the burden from the onset to show with particularity why the patent it challenges is unpatentable." *Harmonic Inc. v. Avid Tech., Inc.*, 815 F.3d 1356, 1363 (Fed. Cir. 2016) (citing 35 U.S.C. § 312(a)(3) (requiring *inter partes* review petitions to identify "with particularity . . . the evidence that supports the grounds for the challenge to each claim")). This burden of persuasion never shifts to the patent owner. *See Dynamic Drinkware, LLC v. Nat'l Graphics, Inc.*, 800 F.3d 1375, 1378 (Fed. Cir. 2015). Moreover, a petitioner should not "place the burden on [the Board] to sift through information presented by the Petitioners, determine where each element [of the challenged claims] is found in [the cited references], and identify any differences between the claimed subject matter and the teachings of [the cited references.]" *Google Inc. v. EveryMD.com LLC*, IPR2014-00347, Paper 9 at 25 (PTAB May 22, 2014).

Anticipation is a question of fact, as is the question of what a prior art reference teaches. *In re NTP, Inc.*, 654 F.3d 1279, 1297 (Fed. Cir. 2011). "Because the hallmark of anticipation is prior invention, the prior art reference—in order to anticipate under 35 U.S.C. § 102—must not only disclose all elements of the claim within the four corners of the document, but must also disclose those elements 'arranged as in the claim.'" *Net MoneyIN, Inc. v. VeriSign, Inc.*, 545 F.3d 1359, 1369 (Fed. Cir. 2008) (quoting *Connell v. Sears, Roebuck & Co.*, 722 F.2d 1542, 1548 (Fed. Cir. 1983)). Whether a reference anticipates a claim is assessed from the skilled artisan's perspective. *See Dayco Prods., Inc. v. Total Containment, Inc.*, 329 F.3d 1358, 1368 (Fed. Cir. 2003) ("[T]he dispositive question regarding anticipation [i]s whether one skilled in the art would reasonably understand or infer from the [prior art reference's] teaching that every claim element was disclosed in that single reference." (quoting *In re Baxter Travenol Labs.*, 952 F.2d 388, 390 (Fed. Cir. 1991))).

A claim is unpatentable under 35 U.S.C. § 103 if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. *See KSR Int'l Co. v. Teleflex Inc.*, 550 U.S. 398, 406 (2007). The question of obviousness is resolved on the basis of underlying factual determinations including: (1) the scope and content of the prior art; (2) any differences between the claimed subject matter and the prior art; (3) the level of ordinary skill in the art; and (4) objective evidence of nonobviousness, if any. *Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966).

In analyzing the obviousness of a combination of prior art elements, it can be important to identify a reason that would have prompted one of skill in the art "to combine . . . known elements in the fashion claimed by the patent at issue." *KSR*, 550 U.S. at 418. A precise teaching directed to the specific subject matter of a challenged claim is not necessary to establish obviousness. *Id.* Rather, "any need or problem known in the field of endeavor at the time of invention and addressed by the patent can provide a reason for combining the elements in the manner claimed." *Id.* at 420.

Accordingly, a party that petitions the Board for a determination of unpatentability based on obviousness must show that "a skilled artisan would have been motivated to combine the teachings of the prior art references to achieve the claimed invention, and that the skilled artisan would have had a reasonable expectation of success in doing so." *In re Magnum Oil Tools Int'l, Ltd.*, 829 F.3d 1364, 1381 (Fed. Cir. 2016) (internal quotation marks omitted). "Both the suggestion and the expectation of success must be founded in the prior art, not in the applicant's disclosure." *In re Dow Chemical Co.*, 837 F.2d 469, 473 (Fed. Cir. 1988).

An obviousness analysis "need not seek out precise teachings directed to the specific subject matter of the challenged claim, for a court can take account of the inferences and creative steps that a person of ordinary skill in the art would employ." *KSR*, 550 U.S. at 418; *see In re Translogic Tech, Inc.*, 504 F.3d 1249, 1259 (Fed. Cir. 2007). In *KSR*, the Supreme Court also stated that an invention may be found obvious if trying a course of conduct would have been obvious to a POSITA:

> When there is a design need or market pressure to solve a problem and there are a finite number of identified, predictable solutions, a person of ordinary skill has good reason to pursue the known options within his or her technical grasp. If this leads to the anticipated success, it is likely the product not of innovation but of ordinary skill and common sense. In that instance the fact that a combination was obvious to try might show that it was obvious under § 103.

550 U.S. at 421. "*KSR* affirmed the logical inverse of this statement by stating that § 103 bars patentability unless 'the improvement is more than the predictable use of prior art elements according to their established functions.'" *In re Kubin*, 561 F.3d 1351, 1359–60 (Fed. Cir. 2009) (citing *KSR*, 550 U.S. at 417).

We analyze the asserted grounds of unpatentability in accordance with these principles to determine whether Petitioner has met its burden to show by a preponderance of the evidence that the challenged claims are unpatentable.

B. Summary of the Cited Prior Art

1. Lesniak (Ex. 1003)

Lesniak is titled "Non-Greasy Personal Care Compositions." Ex. 1003, code (54). Lesniak provides an anhydrous personal care composition, comprising

> (a) at least one ester, wherein said at least one ester is selected from one or more of the following groups: (i) triglycerides, (ii) diglycerides, (iii) monoglycerides, (iv) monoesters of diols, and (v) diesters of diols, and wherein a total concentration of said at least one ester is 45 to 60 weight %, based on the total weight of the composition; (b) a first wax having a melting point of 35° C. to 72° C. present in an amount of 5 to 25 weight %, based on the total weight of the composition; (c) a second wax having a melting point of 73° C. to 90° C. present in an amount of 0.5 to 15 weight% based on the total weight of the composition; (d) at least one plant oil having a saponification value of 150 to 275 mg KOH/g, wherein a total concentration of said at least one plant oil is from 5 to 35 weight%, based on the total weight of the composition.

Id. at 5:32–48. Lesniak describes that "[o]ptionally, the at least one ester is a combination of caprylic triglyceride and capric triglyceride" in ranges from

50–60 weight %, based on the total weight of the composition. *Id.* at 2:38–39, 3:42–44. "Optionally, the composition further comprises zinc oxide . . . present in a concentration of 10 to 30 weight %." *Id.* at 3:39–42. Lesniak describes that the composition may be in the form of a stick. *Id.* at 4:22. According to Lesniak, the compositions possess "good micro-robustness, . . . reduced greasiness and an improved after-feel (such as reduced tackiness and drag) on the skin as compared to other anhydrous compositions[, and] . . . are also physically stable at high and low temperatures." *Id.* at 5:50–54.

Lesniak discloses a desire to "to formulate such compositions from all-natural ingredients (i.e. '100% natural' compositions)." *Id.* at 5:29–31. Lesniak discloses that "the term 'natural compositions' indicates that the composition is formulated from ingredients which are extracted from the earth or sea, derived from plants, or synthesized by other organisms." *Id.* at 4:61–64. Lesniak discloses that the composition may comprise "essential oils which may be used as natural fragrances." *Id.* at 8:38.

2. Phinney (Ex. 1004)

Phinney is titled "Deodorant Formulation." Ex. 1004, code (54). Phinney relates to a deodorant "formulation and preparation thereof with natural ingredients in the absence of heavy metal complexes and/or petroleum based or related compounds." Ex. 1004, 1:7–9. Phinney discloses finding that "a mixture of the hydroxides of zinc and magnesium could provide long lasting protection against odour due to perspiration . . . and are dispersible in aerosols, moisturizing creams, sticks, sprays and roll-on type applicators." *Id.* at 1:17–22.

Phinney explains that zinc and aluminum oxide are "very unreactive against perspiration odour due to insolubility and low hydrolysis rate." *Id.* at 2:35–36. According to Phinney, "magnesium hydroxide, . . . has low solubility and therefore does not 'wash away' under heavy perspiration" and thus "provides 16–20 hour protection and more for a broad cross section of people." *Id.* at 5:47–51. Phinney discloses an embodiment of a specific formulation prepared as a solid stick that "includes magnesium oxide in an amount of between 1% and 2.5% by weight and zinc oxide in an amount of between 0.5% and 1.5% by weight." *Id.* at 6:32–34, 6:12, 8:46–57 (claim 1).

3. Lamb (Ex. 1005)

Lamb is an article titled "Sodium Bicarbonate: An Excellent Deodorant" that was published in *The Journal of Investigative Dermatology*. Ex. 1005, 131. Lamb describes a study that evaluated the effectiveness of baking soda as an underarm deodorant in 90 patients. *Id.* at 131–33. Lamb hypothesizes that the perspiration's free fatty acids, which possess a disagreeable odor, forms a less malodorous sodium salt in the presence of sodium bicarbonate. *Id.* at 133. Based on the results of the study, Lamb concludes that "bicarbonate of soda is a valuable underarm deodorant for common daily usage." *Id.* Lamb observed that "[n]o furunculosis," or boils, "developed in over 90 cases which have used bicarbonate of soda over periods of from four weeks to a year's duration." *Id.*

4. Bianchi '518 (Ex. 1006)

Bianchi '518 is titled "Low Residue Solid Antiperspirant." Ex. 1006, code (54). Bianchi '518 "relates to antiperspirant sticks, substantially free of water and long chain fatty alcohols, which provide the user with excellent antiperspirant efficacy, reduced residue when the composition is first applied to the skin, reduced residue after dry down, high temperature stability, and excellent cosmetics and aesthetics. *Id.* at 1:8–13.

Bianchi '518 discloses that the antiperspirant composition possesses a specification of "Penetration-ASTM D-1321@ 100/77/5 10–16." *Id.* at 2:57–58. Bianchi '518 explains that

> [o]ne important physical parameter affecting consumer acceptance of suspensoid antiperspirant sticks of the invention is hardness as measured by needle penetration. Typical compositions of the invention will range in hardness from about 7.0 to about 14.0 mm as shown by the above mentioned needle penetration test. Products outside this range will typically generate low consumer acceptance ratings.

Id. at 6:23–29.

5. Native (Ex. 1007)

Native is a printed publication from a website advertising a deodorant product sold in 2016. Ex. 1007, 5–8, 11–12, 14–18. Native is advertised as a "Deodorant That Isn't A Chemistry Experiment," which contains "Natural Ingredients" and is "Aluminum-Free." *Id.* at 5. Native lists each ingredient for the Native deodorant stick product, including, *inter alia*, shea butter, beeswax, baking soda, and stearyl alcohol. *Id.* at 6, 11.

The picture on page 5 of Native, reproduced below, illustrates a deodorant product. *Id.* at 5.

The picture of the Native deodorant product, reproduced above, illustrates the deodorant stick in a supine position with the cap removed. *Id.*

6. *Bianchi '254 (Ex. 1008)*

Bianchi '254 is titled "Antiperspirant Stick Compositions." Ex. 1008, code (54). Bianchi '254 discloses "antiperspirant stick compositions comprising specified amounts of antiperspirant active; carrier oil comprising volatile silicone oil and non-volatile masking oil; structurant comprising fatty alcohol and cosmetically acceptable wax having a melting point of 75 to 95° C., a portion of wax comprises polyethylene in specified amounts." *Id.* at code (57). Bianchi '254 addresses

> a need for non-whitening fatty alcohol-containing antiperspirant sticks, particularly sticks containing fatty alcohol at levels of 12% by weight or above, more particularly, 15 wt % or above. Also desired are non-whitening fatty alcohol-containing sticks that have desirable antiperspirant payouts, while providing a 'clean', non-sticky feel. There also remains a need for a non-whitening fatty-alcohol containing stick that provides desirable payout as well as being thermally stable and resistant to dome collapse at temperatures of 50° C.

Id. at ¶ 10. Table 5 of Bianchi '254 provides eight different formulations of antiperspirant sticks with each of their respective components and amounts listed. *Id.* at ¶ 131.

Bianchi ’254 explains that the “[t]he hardness of the sticks can be measured in a needle penetration test.” *Id.* at ¶ 26. Bianchi ’254 describes the method for measuring stick hardness using a “brass/steel tapered needle (DIN 51-579)” depth on six locations of the stick, “with the hardness values being reported as an average of the six measurements.” *Id.* at ¶ 117.

7. Easy Homemade (Ex. 1009)

Easy Homemade is a printed publication of a blog post titled “Easy Homemade Deodorant Recipe with just 3-Ingredients.” Ex. 1009, 5. Easy Homemade describes the blogger’s first experience with natural deodorant, but falling “victim to the burning fiery rash you get after using deodorant with baking soda.” *Id.* at 6. Easy Homemade describes that the blogger discovered that she “could swap the baking soda with magnesium hydroxide - basically the same thing as Milk of Magnesia - and it worked just as well.” *Id.* The Easy Homemade blogger describes melting “two natural deodorants” that are aluminum-free, and making a “[h]uge difference” in each deodorants’ performance by adding “arrowroot powder and magnesium hydroxide to them.” *Id.* at 7. Easy Homemade provides a deodorant recipe and lists “2.15 oz. natural aluminum free deodorant (base) of choice[,] 1 oz. arrowroot powder[,] .5 oz. magnesium hydroxide[,]” to which one may “[a]dd essential oils if desired.” *Id.* at 7–8.

8. Millet (Ex. 1044)

Millet is titled “Anhydrous Deodorant Composition Made from Bicarbonate.” Ex. 1044, code (54). Millet “relates to an anhydrous cosmetic composition made from bicarbonate salt, []the cosmetic use of same as a deodorant,” and “a deodorant product containing said composition.” *Id.* at code (57). Millet describes that it is “possible to formulate anhydrous deodorants . . . in the form of sticks of soft texture, based on ingredients of natural origin," that "have proved easy to apply, effective against perspiration odors and capable of depositing a soft, non-greasy and non-tacky film on the skin." *Id.* at ¶ 4. Millet describes adding "pulverulent fillers, which are suitable for absorbing moisture and sweat," such as, "in particular []native starch, []modified starch and cellulose." *Id.* at ¶¶ 38, 39; *see also id.* ¶¶ 48–50 (disclosing two examples of a "Deodorant Balm" and each of their respective ingredients and percentage amounts.).

9. Anconi (Ex. 1057)

Anconi is titled "Antiperspirant Compositions." Ex. 1057, code (54). Anconi discloses deodorant formulations comprising zinc salts that are substantially free of aluminum salts. *Id.* at Abstract. Anconi also discloses triethyl citrate as an "ingredient[] known to have antibacterial and deodorant properties." *Id.* at ¶ [0027].

10. Stoia (Ex. 1058)

Petitioner's Exhibit 1058 is a report titled "Selected Evidence-Based Health Benefits of Topically Applied Sunflower Oil," which purports to "summarize the outcomes of *in vivo* and *in vitro* studies relating to cosmeceutical and phytopharmaceutical potency of sunflower seed oil based on the epidemiological evidence published" prior to 2015. Ex. 1058, Abstract.

11. Squatch (Ex. 1059)

The reference Petitioner identifies as "Squatch" is a declaration by Vinod Stephen Pinto, Senior Director of Product Management, Ecommerce at Dr. Squatch. Ex. 1059 ¶ 1. Mr. Pinto testifies that images provided in

Exhibit A of the declaration are images of certain Dr. Squatch products that were published on the https://drsquatch.com website in 2020.

12. Sturgis (Ex. 1073)

Sturgis is titled "Deodorant Compositions." Ex. 1073, code (54). Sturgis discloses using at least about 25% of one or more liquid triglyceride as a structurant in a deodorant stick. *Id.* at ¶¶ [0005], [0042]–[0043]. Sturgis also describes that its deodorant sticks "having a hardness from about 80 mm*10 to about 140 mm*10, as measured by penetration with ASTM D-1321 needle" so as to "offer[] a pleasant consumer experience." *Id.* at ¶¶ [0005], [0024]. Sturgis discloses that a starch such as tapioca starch or arrowroot powder can be added to deodorant sticks "for dry feel or wetness absorption." *Id.* at ¶ [0065], Table 1.

C. The Lesniak Grounds (Ground 1–4)

In Grounds 1–4, Petitioner asserts that claims 1–19 would have been obvious over Lesniak in combination with Bianchi '254, Phinney, Anconi, and Easy Homemade. In Ground 1, Petitioner asserts that Lesniak teaches the majority of the limitations of claims 1–13 and relies on Bianchi '254 for its teaching that stearyl alcohol is a fatty alcohol widely used as a structurant for deodorants and for teaching that the claimed hardness range of 8.0 mm to about 14.0 mm[1] was standard and important for consumer acceptance. Pet. 16–26. In Ground 2, Petitioner relies on Phinney for its teaching magnesium hydroxide as an effective deodorant. *Id.* at 26–29 (claim 14). In Ground 3, Petitioner relies on Anconi teaching the use of triethyl citrate as

[1] The claimed hardness range of about 80 mm*10 to about 140 mm*10 is equivalent to 8.0 mm to about 14.0 mm. *See* Ex. 1001, 12:52–55 (explaining that 80 mm*10 is the same as 8 mm).

an antimicrobial. *Id.* at 29–35 (claims 15, 16, 18, 19). In Ground 4, Petitioner relies on Easy Homemade for its teaching on the utility of arrowroot as a starch in deodorants. *Id.* at 35–38 (claim 17). Petitioner also provides detailed claims charts and/or discussion for each of these grounds explaining how each claim limitation is disclosed in the asserted references and how the claims recite deodorants formulated by combining well-known deodorant ingredients with design choices Petitioner asserts would have been obvious. *See* Pet. 20–26 (Ground 1), 28–29 (Ground 2), 32–34 (Ground 3), 38 (Ground 4). Petitioner supports its assertions with the testimony of Dr. Michniak-Kohn. Ex. 1002 ¶¶ 58–134.

After considering all of the arguments and cited evidence of record, we determine that preponderance of the evidence supports the conclusion that claims 1–19 would have been obvious over Lesniak in combination with Bianchi '254, Phinney, Anconi, and/or Easy Homemade. Our reasoning follows.

1. Ground 1 (Claims 1–13)

a) Independent Claim 1

Petitioner argues that claims 1–13 are obvious over Lesniak in view of Bianchi '254. Pet. 16–26. Specific to independent claim 1, Petitioner contends that Lesniak describes natural, anhydrous deodorant sticks containing liquid triglycerides (e.g., caprylic/capric triglyceride), structurants (e.g., beeswax, fatty alcohols), and antimicrobials (e.g., zinc oxide). Pet. 16–17, 20–22. Petitioner further contends that:

> While Lesniak describes formulating deodorant compositions with "fatty alcohols," which a POSA would have understood to include stearyl alcohol, it does not expressly disclose using stearyl alcohol as a structurant. Ex. 1003, 8:31-32; Ex. 1001,

> 5:43-44, 61-62; Ex. 1008, Tables 1, 5. Bianchi ’254 states stearyl alcohol is a “widely used” structurant “of particular interest” for deodorants and describes examples comprising 12-25% of stearyl alcohol. Ex. 1008, ¶¶[0001], [0018], [0019], [0058]. Therefore, a POSA would have found it obvious to use stearyl alcohol as a structurant in the composition of Lesniak, as taught by Bianchi ’254.

Pet. 17. Petitioner also contends that Lesniak’s focus on “natural” formulations suggests the absence of aluminum salts, baking soda, and synthetic silicones/fragrances. *Id.* at 21–23. For example, Petitioner contends that the use of zinc oxide in Lesniak serves as a known alternative to aluminum and Lesniak’s preference for natural ingredients militates against using aluminum, which some consumers perceive as unsafe. *Id.* at 21–22.

Petitioner acknowledges that Lesniak does not disclose a hardness from about 80 mm*10 to about 140 mm*10. *Id.* at 17. For this element, Petitioner relies on Bianchi ’254, which describes preferred hardness values (8 to 15 mm) measured by penetration with a standard ASTM-D 1321 needle. *Id.* at 18. Petitioner contends that this hardness range was standard and that it would be obvious to formulate Lesniak’s deodorant stick to be within the claimed hardness range, which overlaps with ranges known to be acceptable to consumers. *Id.* at 18, 22, (citing Ex. 1002 ¶¶ 63–65 (citing Ex. 1025, 12:41–55), 75–77). Petitioner contends that adjusting the amount of structurant to reach a target hardness (e.g., adjusting amounts to attain the desired feel of the stick) would have been a matter of routine optimization for a formulator. Pet. 8 (citing Ex. 1041, 4:37–44; Ex. 1002 ¶¶ 45–47). For example, “[a] POSA would have understood that the hardness could be adjusted by altering the amount of the [primary] structurant and/or by incorporating secondary structurants." *Id.* at 20 (citing Ex. 1002 ¶¶ 66–67).

Patent Owner responds to Petitioner's challenge of claim 1 with several arguments, which we address below. PO Resp. 14–40.

(1) Whether Lesniak discloses a deodorant stick

Patent Owner first contends that "Lesniak does not disclose a 'deodorant stick,' as required by the preamble of every independent claims." PO Resp. 14. Patent Owner acknowledges that Lesniak suggests that "[t]he composition can also be an antiperspirant" (Ex. 1003, 10:27–29), but contends that Lesniak fails to "specify any form of antiperspirant, which can come in a variety of forms, including aerosol sprays, creams, liquids, and gels." PO Resp. 18 (citing Ex. 2011 ¶ 57). Patent Owner further contends that Lesniak's "passing references to 'antiperspirant' fail to convert Lesniak into something other than a reference about sunscreens and diaper creams." *Id.*

We are not persuaded by Patent Owner's arguments. Rather, we agree with Petitioner that Lesniak teaches anhydrous personal care compositions where "[t]he composition can . . . be an antiperspirant" and "can also be a stick." Ex. 1003, 9:45, 10:27. In light of those express disclosures, we are unpersuaded by Patent Owner's arguments that Lesniak fails to teach or suggest a "deodorant stick." We thus determine that Lesniak teaches or suggests a deodorant in the form of an antiperspirant. Pet. Reply 2–3 (citing Ex. 1036, 2 ("all antiperspirants are deodorants"); Ex. 1037, 246 (same); Ex. 1084, 33 ("[T]he main mechanisms through which deodorant products intended to combat body odors . . . were the usage of antiperspirants that prevent perspiration[.]")).

(2) Whether antiperspirants must contain aluminum

Patent Owner contends that Lesniak's antiperspirants must contain aluminum. PO Resp. 19–20. Specifically, Patent Owner contends that,

> [T]he term "antiperspirant," to a POSA, means a composition containing an aluminum salt or complex—the only antiperspirant actives approved by the FDA. *See* 37 C.F.R. [§] 350. As Dr. Lochhead explains, a POSA would have understood that the FDA regulates antiperspirants as drugs and regards deodorants as cosmetics. (Ex. 2011 at ¶ 52.) A POSA would have further understood Lesniak's disclosure of "antiperspirant" as requiring the use of an antiperspirant active, and the only antiperspirant actives approved by the FDA all contain aluminum. (*Id.*) Indeed, the FDA lists the 18 approved antiperspirant actives in its monograph "Antiperspirant Drug Products for Over-the-Counter Human Use," and every single one comprises an aluminum compound or complex.

PO Resp. 19–20 (footnote omitted). *See also* PO Sur-reply 18–20 ("a POSA's understanding of 'antiperspirants,' now and at the time of Lesniak, is that they contain aluminum" (citing Ex. 1081,[2] 60:12–14 ("[A] POSA would understand if you're talking about an antiperspirant, it has to have aluminum."); Ex. 2011 ¶¶ 52–54)).

We are not persuaded by Patent Owner's argument. Rather, we agree with Petitioner that, while Patent Owner "emphasizes that the FDA requires antiperspirants marketed in the U.S. to contain aluminum, anticipation and obviousness are distinct from FDA regulatory requirements." Pet. Reply 3 (citing *Persion Pharms. LLC v. Alvogen Malta Ops. Ltd.*, 945 F.3d 1184, 1193 (Fed. Cir. 2019) ("FDA's approval requirements do not undermine the force of the evidence as to obviousness")). Further to that point, the evidence of record shows that zinc compounds were known antiperspirant

[2] Ex. 1081, Dr. Robert Lochhead Deposition Transcript, July 10, 2025.

actives and alternatives to aluminum in antiperspirant compositions. *Id.* at 4 (citing Ex. 1003, 3:39–4:21, 8:46–48, 10:27 (describing zinc-based antiperspirant compositions); Ex. 1014 ¶¶ 20–26 ("[A]ntiperspirant actives include astringent metallic salts, particularly inorganic and organic salts of aluminum, zirconium, and zinc, as well as mixtures thereof.")); *see also* Ex. 1002 ¶ 74 ("Lesniak's formulations include zinc oxide, which was a known alternative to aluminum.") (citing Ex. 1003, 1:39–2:30, 3:43–4:21, 5:32–6:23, 8:31–9:59).

(3) Whether Lesniak enables a deodorant stick

Patent Owner contends that Lesniak fails to provide sufficient guidance as to how to formulate a stable deodorant or antiperspirant stick, and therefore, is non-enabling. PO. Resp. at 23–29 (citing Ex. 2011 ¶¶ 57–69). In particular, Patent Owner contends that "the only 'stick' formulation Lesniak expressly teaches is a sunscreen stick." *Id.* at 23–24 (citing Ex. 1003, Table 7). Patent Owner also directs our attention to the results disclosed in Lesniak's [Table 2] and argues that Lesniak's compositions suffered from stability issues. *Id.* at 24–28; *see also* Sur-reply 17 ("Lesniak's Table 2 shows that even small changes in a composition can yield unstable results," demonstrating that the art is unpredictable.). Patent Owner concludes that "Lesniak provides no teaching of a stable 'deodorant stick,' no teaching of a stable 'antiperspirant stick,' and at least nine different sunscreen formulas with slight variations that are all unstable." PO Resp. 28 (citing Ex. 2011 ¶ 69).

We find that the evidence of record fails to support Patent Owner's argument that the prior art does not enable a stable deodorant stick. Rather, we are persuaded by Petitioner that Lesniak sufficiently teaches compositions with the relevant ingredients and achieves stable compositions through the use of a second wax having a melting point of 73 to 90° C. Pet. Reply 6–7 (citing Ex. 1003, 1:39–54). For example, Lesniak's compositions (nos. 8 and 11) that used such a secondary wax, namely CastorWax MP80, are disclosed as sufficiently stable. Ex. 1003, Table 2. Here, we further credit the testimony of Dr. Michniak-Kohn that

> Lesniak teaches how to make stable compositions by including a secondary wax, such as castor wax, with a sufficiently high melting point. Ex. 1003, 1:39-54, 5:23-32, Table 2. Both of the compositions (nos. 8 and 11) tested by Lesniak that had such a wax passed the stability tests. *Id.*

Ex. 1086 ¶ 7; *see also* Pet. Reply 6 ("Lesniak overcame any stability issues, as it expressly discloses compositions that are 'physically stable at high and low temperatures.'") (citing Ex. 1003, 5:23–32).

Having considered the parties' positions and evidence of record, summarized above, we determine that Lesniak provides sufficient disclosure to enable a person of ordinary skill in the art to formulate deodorant stick within the parameters of claim 1.

(4) Motivation to Combine Lesniak and Bianchi '254

Patent Owner asserts that Petitioner's argument is inconsistent because a POSA would have understood Lesniak's formulations to be complete deodorant sticks that did not require additional ingredients, yet Petitioner seeks to modify Lesniak's formulations. PO Resp. 29–31 (citing Pet. 20–26, 29–31). Patent Owner compares the formulations of Lesniak and Bianchi '254 asserting that a "POSA would not have been motivated to consider Bianchi-254's teachings in combination with Lesniak, as the Bianchi compositions require large amounts of non-natural ingredients (volatile silicones), and using such non-natural ingredients would defeat the core objective of Lesniak to provide natural compositions." PO Resp. 30–31.

We have considered Patent Owner's arguments but determine that Petitioner has the better position. Petitioner does not rely on the teachings of Bianchi '254 relating to its non-natural ingredients, such as silicones, but on its teaching concerning an acceptable hardness range to accommodate consumer preferences for deodorant sticks. *See* Pet. 17–18; Ex. 1008 ¶ 26. To that point, Lesniak teaches the use of a first and second wax as preferred structurants and the evidence of record supports a conclusion that a POSA would understand how to formulate a deodorant stick using the structurants taught in Lesniak. *See*, *e.g.*, Pet. 18–23; Ex. 1002 ¶ 42 ("For deodorant sticks, the structurant is what [is] mixed with the carrier oil to cause the formulation to have the consistency (e.g., hardness) of a stick form. In addition to adjusting the balance between, e.g., waxes of different melting points to control hardness, a POSA would have appreciated that the amount of structurant controls the hardness of the stick."). Therefore, Patent Owner's argument concerning the lack of a reason to combine the references based on the natural or non-natural nature of the ingredients of Lesniak and Bianchi '518 is unpersuasive.

(5) Whether Lesniak or Bianchi '254 Disclose a Stick of the Claimed Hardness

Claim 1 requires that the claimed deodorant stick to have "a hardness from about 80 mm*10 to about 140 mm*10, as measured by penetration with ASTM D-1321 needle." Ex. 1001, 15:25–34. For this element, Petitioner relies on Bianchi '254's teaching that consumers prefer products falling within a hardness range of about 8 mm to about 15 mm (equivalent to about 80 mm*10 to about 150 mm*10), as measured by the standard ASTM-D-1321 needle penetration test. Pet. 17–23 (citing Ex. 1008 ¶¶ 26, 117; *see also* Ex. 1001, 12:52–55 (explaining that 80 mm*10 is the same as 8 mm).

Patent Owner contends that the standard hardness test referenced in Bianchi ’254 is different from the test described in the ’752 patent and contends that “the Bianchi hardness test uses a needle and shaft with *double the weight* of the needle and shaft used in the ’752 Patent [100 grams vs. 50 grams], and thus its so-called same or similar hardness ranges reflect that the Bianchi compositions are actually *significantly harder* than those claimed in the ’752 Patent (*i.e.*, [the ’752 method would] require twice the weight of the penetration needle assembly to yield similar penetration depths).” PO Resp. 32 (citing Ex. 2011 ¶¶ 76–84; Ex. 1087,[3] 461–62 (“Place a 50-g weight above the penetrometer needle, making a total load of 100 ± 0.015 g for the needle and all attachments.”)). Patent Owner supports its argument with the testimony of Dr. Lochhead, who explains that

> The ’752 Patent uses a different hardness testing method. The ’752 Patent explains that its invention exhibits a hardness in the range of “80 mm*10 to about 140 mm*10,” as measured using an ASTM D-1321 penetration needle in a test similar to ASTM D-1321 and DIN 51-579, but with a significant difference: “the total weight of the shaft and needle is 50±0.2 grams when the shaft is in free fall.” (’752 Patent at 12:7-8.) The ’752 Patent’s method does not include the “50-g weight” added to the needle in the DIN 51-579 test used by Bianchi-254.

[3] Ex. 1087, ASTM D-1321 Needle Penetration Test Standard (1996).

Ex. 2011 ¶ 81; *see* PO Resp. 33. Patent Owner further contends that the evidence of record fails to support a conclusion that Bianchi '254's test and the '752 patent's test would produce overlapping hardness ranges. *Id.* at 34.

We find that the evidence of record fails to support Patent Owner's position that the '752 patent discloses a unique hardness test. As noted by Petitioner, the '752 patent discloses that hardness was measured by "penetration with ASTM D-1321 needle (see Hardness test method below)." Ex. 1001, 9:23–24. The '752 patent describes that its hardness test was based on ASTM D-1321 and used penetration "[n]eedles as specified for ASTM D 1321/DIN 51 579." Ex. 1001, 11:61–62. The disclosed Hardness test method appears to correspond to the ASTM D-1321 standard protocol and does not expressly disclose diverting from the standard protocol in any manner, including the '752 patent's disclosure that "[p]eriodic calibration should confirm: . . . the total weight of the shaft and needle is 50±0.2 grams when the shaft is in free fall." *Id.* at 12:2–7. That disclosure for periodic calibration corresponds with the requirement under the ASTM D-1321 test that "[t]he final weight of the needle shall be 2.5 ± 0.05 g. The total weight of the plunger shall be 47.5 ± 0.05 g." Ex. 1087, 461. For the actual test, the ASTM D-1321 standard protocol uses 100 grams of weight. *Id.* at 461–62. That is, the ASTM D-1321 test makes clear that a 50-gram weight is placed "above the penetrometer needle, making a total load of 100 ± 0.15 g for the needle and all attachments," thereby explaining Bianchi '254's disclosure of a test using 100 grams of weight. *Id.*; *see also* Pet. Reply 19; PO Sur-reply 4–5; Ex. 2011 ¶¶ 79–82.

The '752 patent does not disclose performing the test with only the needle as Patent Owner contends. PO Resp. 32. Rather, the '752 patent discloses a hardness test that significantly corresponds to the ASTM D-1321 standard protocol, the test that is expressly reference in the ’752 patent and recited in the claims. The ’752 patent does not expressly disclose any departure from the standard test, including failing to expressly disclose performing the hardness test without the required 50-gram weight placed above the penetrometer needle, and its silence to that fact is insufficient to persuade us that a different test was ultimately intended or performed. *See Thorner v. Sony Computer Ent. Am. LLC*, 669 F.3d 1362, 1365 (Fed. Cir. 2012) (“‘To act as its own lexicographer, a patentee must ‘clearly set forth a definition of the disputed claim term” other than its plain and ordinary meaning.”. . . “[T]he patentee must ‘clearly express an intent’ to redefine the term.” (quoting *CCS Fitness, Inc. v. Brunswick Corp.*, 288 F.3d 1359, 1366 (Fed. Cir. 2002); *Helmsderfer v. Bobrick Washroom Equip., Inc.*, 527 F.3d 1379, 1381 (Fed. Cir. 2008)). Rather, we are persuaded by the following testimony of Dr. Michniak-Kohn:

> Dr. Lochhead states that he believes the claimed recitation of a hardness range “as measured by penetration with ASTM D-1321 needle” should not be understood to indicate the ASTM D-1321 standard, but a modification of that standard. Ex. 2011, ¶81. As a basis for his opinion. Dr. Lochhead focuses on language in the ’752 Patent that “the total weight of the shaft and needle is 50±0.2 grams.” Ex. 2011, ¶25. But he omits the preceding language, which states that this is the weight to be used for “[p]eriodic calibration” to “confirm” that the equipment is in proper working order. Ex. 1001, 12:1-11. That “[p]eriodic calibration” section precedes a section identifying what needs to be confirmed “[a]t the time of use.” *Id.*, 12:10-12. Those two sections would suggest to a POSA routine checking of equipment. Only after those two sections does the specification set forth the test. *Id.*, 12:20-57 (“Sample Preparation and Measurement”). Moreover, the “[p]eriodic calibration” comports

> with the ASTM D-1321 standard, which likewise requires that the "weight of the needle shall be 2.5 g" and the "weight of the plunger shall be 47.5 g," for a 50-g total weight of the needle assembly. Ex. 2018, 5. Although the ASTM D-1321 standard further describes using an additional 50-g weight during the test, a POSA would not read the '752 Patent to exclude that weight. Rather, a POSA would consult the details of the ASTM D-1321 standard identified by the '752 Patent to ensure compliance

Ex. 1086 ¶ 3.

Having considered the parties' positions and evidence of record, summarized above, we determine that a POSA would have been motivated to formulate Lesniak's deodorant stick to be within the claimed hardness range, which overlaps with ranges known to be acceptable to consumers as disclosed in Bianchi '254, with a reasonable expectation of success. Pet. 17–22 (citing Ex. 1002 ¶¶ 63–67, 76–77).

(6) Whether a POSA would have been motivated to use stearyl alcohol as the structurant of the stick

Petitioner contends that Bianchi '254 discloses stearyl alcohol as a structurant widely used for deodorants and describes examples including 12–25% stearyl alcohol as the structurant present in the greatest amount. Pet. 17 (citing Ex. 1008 ¶¶ 1, 18, 19, 58). Petitioner further contends that

> A POSA would have been motivated to use stearyl alcohol as a structurant in Lesniak's deodorant formulations and would have had a reasonable expectation of success in doing so because stearyl alcohol was a preferred option that had already been used successfully as a structurant in deodorant sticks.

Id. at 21 (citing Ex. 1002 ¶ 72).

Patent Owner contends that "the POSA would not have been motivated to reformulate Lesniak, and instead understood Lesniak as 'a complete deodorant stick' without the need for additional ingredients or modification." PO Resp. 39. Patent Owner further contends that POSA would not have had a reasonable expectation of success in reformulating Lesniak to make stearyl alcohol the primary structurant given Lesniak's stability issues and "the fundamental differences between the compositions described in Lesniak and that of the Bianchi '254." *Id.* at 40.

Having considered the parties' positions and evidence of record, summarized above, we find Petitioner to have the better position, and we conclude that Patent Owner's arguments fail to overcome Petitioner's sufficient showing that a POSA would have been motivated to use stearyl alcohol as the structurant of the stick. To begin, we note that Lesniak discloses the use of a first wax having a melting point of 35° C. to 72° C. and a second wax having melting point of 73° C. to 90° C. Lesniak also discloses that

> such compositions are . . . physically stable (i.e. do not exhibit separation into a bulk solid portion and a bulk liquid portion) when stored at 49° C./0% relative humidity for a period of 13 weeks, and when stored at 0° C./75% relative humidity for a period of 13 weeks, and show acceptable dispensing characteristics in addition to being non-greasy.

Ex. 1003, 5:61–67.

Furthermore, Lesniak teaches that the first wax may be "monoesters of long chain fatty acids and long chain fatty alcohols." *Id.* at 1:66–2:2. Here, Dr. Michniak-Kohn informs us that a POSA would have understood stearyl alcohol to fall within the category of fatty alcohols suggested by Lesniak, a point that is undisputed by Patent Owner. Ex. 1002 ¶ 61. For its part, "Bianchi '254 states stearyl alcohol is a 'widely used' structurant 'of particular interest' for deodorants and describes examples comprising 12–25% of stearyl alcohol" as the structurant present in the greatest amount.

Ex. 1002 ¶ 104 (citing Ex. 1008 ¶¶ 1, 18, 19, 58). In view of those teachings in Lesniak and Bianchi '254, we agree with Petitioner that a POSA would have found it obvious to choose stearyl alcohol as the primary structurant (first wax) in the composition of Lesniak, as taught by Bianchi '254. *See Almirall, LLC v. Amneal Pharms. LLC*, 28 F.4th 265, 273 (Fed. Cir. 2022) (It is obvious to "substituting one known [material] for another. Each may be effective at a different concentration in different formulations, but that is just a property of the particular known material, subject to conventional experimentation."); *see also Conopco Inc. v. The Procter & Gamble Co.*, IPR2013-00505, Paper No. 69, at 23 (PTAB Feb. 10, 2015), *aff'd*, No. 2015–1680 (Fed. Cir. Aug. 8, 2016) ("no express suggestion would have been needed to prompt one to substitute a known, interchangeable alternative for another, to perform a common function").

(7) Conclusion as to Claim 1

For the reasons discussed above, we conclude that Petitioner has proved by a preponderance of evidence that the combined teachings of Lesniak and Bianchi '254 render claim 1 obvious.

b) Dependent Claims 2–13

Petitioner argues that claims 2–13 would have been obvious over Lesniak and Bianchi '254. Pet. 22–26. Patent Owner presents the arguments discussed above regarding the limitations of claim 1, but does not present arguments or direct us to evidence against Petitioner's challenges that are specific to the limitations of claims 2–13. *See generally* PO Resp.

Upon review of Petitioner's undisputed arguments and evidence, which we adopt as our own, and the evidence of record, we determine that Petitioner has demonstrated by a preponderance of the evidence that each of claims 2–13 would have been obvious over the combination of Lesniak and Bianchi ’254. *See*, *e.g.*, Pet. 17–22.

2. *Ground 2: Obviousness over Lesniak, Bianchi ’254, and Phinney (Claim 14)*

Claim 14 depends from claim 1 and adds a requirement that the primary antimicrobial is selected from a group that includes magnesium hydroxide. Petitioner argues claim 14 is obvious by adding Phinney to the combination of Lesniak and Bianchi ’254, discussed above. Pet. 26–29.

Petitioner concedes that Lesniak does not expressly teach the use of magnesium hydroxide. *Id.* at 26. Petitioner contends that “[b]ecause Lesniak teaches using zinc oxide (a known antimicrobial) and Phinney teaches that using magnesium hydroxide in combination with zinc oxide improves antimicrobial properties, a POSA would have had a reason to follow Phinney’s teaching and add magnesium hydroxide to Lesniak’s composition.” *Id.* at 27. Petitioner contends that both Lesniak and Phinney focus on natural ingredients, making the substitution or addition of magnesium hydroxide a logical design choice. *Id.* at 27–28.

Patent Owner contends that Lesniak and Phinney describe entirely different uses for their metal oxides/hydroxides. PO Resp. 41. In particular, Patent Owner contends that Lesniak includes zinc oxide in sunscreen and diaper cream compositions as a protective barrier, not for odor control. *Id.* Moreover, Patent Owner contends that “Lesniak’s formulas with high concentrations of zinc oxide [are] unsuitable for use as an underarm deodorant or antiperspirant.” *Id.* (citing Ex. 2011 ¶ 96; Ex. 1004, 2:33–36 (“[Z]inc oxide… [is] very unreactive against perspiration odour.”), 3:20–21). Thus, Patent Owner asserts a “POSA would not have been motivated to look at Phinney to modify Lesniak or incorporate any additional antimicrobial into Lesniak 'to improve odor protection,' as odor protection was not a goal of Lesniak's zinc oxide in the first place." *Id.*

Furthermore, Patent Owner contends that Phinney does not suggest a combination of zinc oxide and magnesium hydroxide. *Id.* at 34–35. Rather, according to Patent Owner,

> Petitioner cherry picks magnesium hydroxide from Phinney's disclosure of a three- or four-ingredient combination. Phinney explains that to combat odor, its compositions contain a combination of (1) magnesium hydroxide, and (2) zinc hydroxide, and (3) potassium chloride and/or potassium sulfate. (Ex. 1004 at 8:7-31, cl. 1.) There are two significant aspects to this multi-part odor fighting combination: the requirement of zinc hydroxide, not zinc oxide; and the requirement of potassium chloride/sulfate. (Ex. 2011 at ¶ 97.)

PO Resp. 41–42 (emphasis omitted). Patent Owner further contends that Lesniak and Phinney are incompatible, arguing that Lesniak's compositions are anhydrous while Phinney's compositions requires water. *Id.* at 43 (citing 2011 ¶ 101); *see also* Ex. 1004, 4:48–51, 8:40–41 (explaining magnesium oxide and zinc oxide are both mixed with water to become their respective hydroxides, and that they "must be completely converted to their hydroxides or skin irritation will occur"); Sur-reply 19–20.

We have considered Patent Owner's arguments but find, on this record, Petitioner to have the better position, and we conclude that Patent Owner's arguments fail to overcome Petitioner's sufficient showing that a POSA would have been motivated to add magnesium hydroxide to Lesniak's compositions because

> Phinney teaches that: (i) zinc oxide has "low activity" as a "deodorant agent"; (ii) magnesium hydroxide is "an effective

> deodorant"; and (iii) zinc and magnesium compounds have synergistic effects, hence motivating a POSA to add magnesium hydroxide to Lesniak's zinc oxide deodorants. Ex. 1004, 1:13-3:16, 5:61-67.

Pet. Reply 19; *see also* Pet. 29 (arguing that a POSA would have been motivated to add magnesium hydroxide to the composition of Lesniak because zinc oxide is not a particularly efficacious antimicrobial in deodorants when used alone, whereas magnesium hydroxide is an effective deodorant).

We acknowledge that Phinney suggests mixing zinc oxide with water in order to convert zinc oxide to zinc hydroxide. *See*, *e.g.*, Ex. 1004, 1:17–19 ("It was also found that a mixture of the hydroxides of zinc and magnesium could provide long lasting protection against odour due to perspiration."). The evidence of record, however, shows that both magnesium hydroxide and zinc oxide were known options for deodorant antimicrobials, and claim 14 does not require a minimal level of effectiveness such that the combination of magnesium hydroxide and zinc oxide would need to outperform any other combination including the mixture of zinc and magnesium hydroxides. *See e.g.*, Ex. 1003, 3:39–42; Ex. 1004; *see also In re Fulton*, 391 F.3d 1195, 1200 (Fed. Cir. 2004) ("Our case law does not require that a particular combination must be the preferred, or the most desirable, combination[.]"). Here, we also credit the testimony of Dr. Michniak-Kohn that "POSA would understand that the water-based vs. anhydrous distinction is not relevant, as a POSA would understand that magnesium hydroxide would still function as an antimicrobial in anhydrous compositions such as those disclosed by Lesniak." Ex. 1086 ¶ 9 (citing Ex. 1072, 1–2, 5 (disclosing deodorants withoutadded water and with zinc oxide and/or magnesium hydroxide)). Accordingly, we are persuaded by Petitioner that "[a] POSA would have been motivated to incorporate known improvements to odor protection, particularly where the known improvement also was known to maintain the 'natural' composition intended by Lesniak. Pet. 27 (citing Ex. 1002 ¶ 95).

Ultimately, however, it does not matter whether Phinney teaches a composition that contains water. "[I]t is not necessary that [two pieces of prior art] be physically combinable to render obvious" the claimed subject matter. *Allied Erecting and Dismantling Co., Inc. v. Genesis Attachments*, LLC, 825 F.3d 1373, 1381 (Fed. Cir. 2016) (quoting *In re Sneed*, 710 F.2d 1544, 1550 (Fed. Cir. 1983)). Rather, the question is "whether the claimed inventions are rendered obvious by the teachings of the prior art as a whole." *In re Etter*, 756 F.2d 852, 859 (Fed. Cir. 1985) (noting that whether one prior art reference can be incorporated into another is "basically irrelevant.") Lesniak and Phinney undisputedly teach that magnesium hydroxide and zinc oxide were known ingredients in personal care products including deodorants. Ex. 1003, 3:39–42; Ex. 1004. The inclusion of magnesium hydroxide and zinc oxide into a single product therefore represents, absent evidence to the contrary, "a combination of familiar elements according to known methods that does no more than yield predictable results." *Agrizap, Inc. v. Woodstream Corp.*, 520 F.3d 1337, 1344 (Fed. Cir. 2008).

For the reasons discussed above, we conclude that Petitioner has proved by a preponderance of evidence that the combined teachings of Lesniak and Phinney render obvious claim 14.

3. Ground 3: Obviousness over Lesniak, Bianchi ’254, Phinney, Anconi (Claim 15, 16, 18, 19)

Petitioner asserts that the combination of Lesniak, Bianchi ’254, Phinney, and Anconi renders obvious claims 15, 16, 18, and 19. Pet. 29–35. Claims 15–16 and 18–19 recite deodorant sticks comprising a list of certain ingredients, including triethyl citrate and starch.

Petitioner primarily relies on Lesniak to teach the limitations of claims 15, 16, 18, and 19 Petitioner but acknowledges that, “[w]hile Lesniak describes formulating deodorant compositions with antimicrobials such as zinc oxide, it does not expressly teach the use of triethyl citrate.” *Id.* at 29–30. For that element, Petitioner relies on Anconi and contends that Anconi discloses deodorant formulations containing antimicrobials and identifies triethyl citrate as an ingredient with known antibacterial and deodorant properties. *Id.* at 30 (citing Ex. 1057 ¶¶ 8–10, 27; Ex. 1002 ¶¶ 103–104). Petitioner further contends that triethyl citrate was known to prevent the degradation of liquid triglycerides, which are core components of Lesniak’s formulations. *Id.* at 30–31.

Similar to arguments asserted in Ground 1, Patent Owner contends that “[a] POSA would have either recognized that Lesniak alone created a complete and efficacious deodorant stick with zinc oxide as an antimicrobial (Ground 1, Petition at 20-21), or that Lesniak’s zinc oxide was ineffective for deodorants and should include additional antimicrobials such as triethyl citrate (Ground 3, Petition at 29-31; Ex. 2011 at ¶ 110.)” PO Resp. 47. Patent Owner also reiterates that the fragility of Lesniak’s compositions would not have provided a reasonable expectation of success in making the modifications to arrive at the formulations of claims 15–16 and 18–19. *Id.* at

47–48. Patent Owner also points to the different types of formulations for Lesniak and Anconi for a lack of reason to combine the teachings of these references with a reasonable expectation of success. *Id.* at 46.

We have addressed these arguments previously in Ground 1 and found them similarly unpersuasive here. We credit Dr. Michniak-Kohn's testimony supporting Petitioner's positions set forth above. *See* Ex. 1002 ¶¶ 102–140. For example, we credit Dr. Michniak-Kohn's testimony concerning the inclusion of starch in Lesniak's formulation with a reasonable expectation of success because starches such as tapioca/arrowroot had been successfully used in natural deodorants before. *See Id.* ¶ 109. We credit also Dr. Michniak-Kohn's testimony concerning the use of triethyl citrate in Lesniak's formulation to improve the ability to combat odor, and that "[a] POSA further would have been motivated to include triethyl citrate in Lesniak's deodorant compositions because . . . those compositions contain liquid triglyceride, and triethyl citrate was known to prevent the degradation of triglycerides in compositions for topical use." *Id.* ¶ 114.

Accordingly, having considered the parties' positions and evidence of record, summarized above, we determine that a POSA would have been motivated to make the combination of Lesniak, Phinney, Bianchi '254, and Anconi to arrive at the claimed inventions of claims 15, 16, 18, and 19 of the '752 patent with a reasonable expectation of success. Thus, we determine that Petitioner has shown by a preponderance of the evidence that these claims would have been obvious over the combination of Lesniak, Phinney, Bianchi '254, and Anconi.

4. *Ground 4: Obviousness over Lesniak, Bianchi ’254, Phinney, Anconi, and Easy Homemade (Claim 17)*

Claim 17 depends from claim 15 and adds a requirement that the starch is arrowroot powder. Petitioner argues claim 17 would have been obvious by adding Easy Homemade to the combination of Lesniak, Bianchi ’254, Phinney, and Anconi, discussed above. Pet. 35–38. Petitioner contends that Easy Homemade describes natural deodorant formulations specifically using magnesium hydroxide and arrowroot powder. *Id.* at 35. Petitioner argues Lesniak teaches starches generally, and Easy Homemade specifically highlights arrowroot powder for moisture absorption in natural deodorants. *Id.* at 36. Petitioner contends that a POSA would have looked to arrowroot for its known moisture-absorbing and thickening properties, making it an obvious alternative to the starches mentioned in Lesniak. *Id.* (citing Ex. 1002 ¶¶ 142–144).

First, Patent Owner contends that Petitioner has not sufficiently established that Easy Homemade is prior art printed publication. PO Resp. 50–51. We disagree and determine that Petitioner has shown by a preponderance of the evidence that Easy Homemade was publicly accessible. A reference is considered publicly accessible if it is “made available to the extent that persons interested and ordinarily skilled in the subject matter or art, exercising reasonable diligence, can locate it.” *Jazz Pharms., Inc. v. Amneal Pharms., LLC*, 895 F.3d 1347, 1355 (Fed. Cir. 2018). To “determine whether an interested researcher would have been sufficiently capable of finding the reference and examining its contents,” we “must consider all of the facts and circumstances surrounding the disclosure.” *In re Lister*, 583 F.3d 1307, 1312 (Fed. Cir. 2009); *see also In re*

Klopfenstein, 380 F.3d 1345, 1350 (Fed. Cir. 2004) (explaining that a determination of whether a reference qualifies as a printed publication "involves a case-by-case inquiry into the facts and circumstances surrounding the reference's disclosure to members of the public").

Patent Owner's main argument appears to be that Soap Deli News was not indexed. *See* PO Resp. 50–51. As Petitioner points out, however, such evidence is not necessarily required in all cases. Pet. Reply 25 (citing *Voter Verified, Inc. v. Premier Election Soln's*, 698 F.3d 1374, 1380 (Fed. Cir. 2012)). Rather, absent evidence of indexing, testimony indicating that the particular online publication or website on which the reference was published was well-known to the community interested in subject matter of the reference, and that, upon accessing the website, those interested would have found the reference using the website's own search functions can support the ultimate determination that a given reference was publicly accessible. *Id.* at 11–12 (citing *Voter Verified*, 698 F.3d at 1380–81). To that point, Petitioner presents evidence from the Wayback Machine for the Easy Homemade Blogpost, *see* Ex. 1009, 1–2, in addition to three other Soap Deli News posts, *see* Exs. 1017–1019. Dr. Mishniak-Kohn testifies "Easy Homemade was widely accessible, being part of a series of related 'Soap Deli News' articles dedicated to deodorant formulations with significant engagement from individuals globally." Ex. 1002 ¶ 187 (citing Ex. 1009; Exs. 1016–1018). Dr. Mishniak-Kohn also testifies that "a POSA would have appreciated that such websites from independent formulators striving for more natural options present an indication of consumer preferences, which a POSA would desire to meet in formulating new commercial products." *Id.*

We credit Dr. Mishniak-Kohn's testimony here, and determine that Easy Homemade was publicly accessible and qualifies as a printed publication.

Next, Patent Owner responds that a POSA would not have had a reason to modify Lesniak as Petitioner asserts Lesniak is "a complete deodorant stick" and "does not identify any deficiency in Lesniak that would have motivated a POSA to introduce arrowroot powder for 'thickening and absorbing properties.'" PO Resp. 51. We have addressed these arguments previously in Ground 1 and found them similarly unpersuasive here. We also credit Dr. Michniak-Kohn's testimony supporting Petitioner's positions set forth above. *See* Ex. 1002 ¶¶ 141–47. For example, we credit Dr. Michniak-Kohn's testimony concerning the inclusion of arrowroot starch in Lesniak's formulation with a reasonable expectation of success because starches such as arrowroot had been successfully used in natural deodorants before for its "well-known thickening and absorbing." *Id.* ¶ 145 (citing Ex. 1068, 62; Ex. 1016, 5 (discussing "arrowroot powder to help keep you dry")); *see also Almirall*, 28 F.4th at 273 (obvious to "substitute one known" option for another). Accordingly, we determine that Petitioner has shown by a preponderance of the evidence that claim 17 is rendered obvious by the combination of Lesniak, Bianchi '254, Phinney, Anconi, and Easy Homemade.

D. The Native Grounds (Grounds 5–8)

Petitioner presents four grounds that rely on Native to teach the majority of the limitations of the asserted claims in Grounds 5 through 8. Pet. 38–63. For Ground 5, Petitioner asserts that claims 1 and 4–14 of the '752 patent would have been obvious over Native and Bianchi '254. Pet. 38–50. For Ground 6, Petitioner asserts that claims 2–3 would have been obvious over Native, Bianchi ’254, and Easy Homemade. Pet. 50–54. For Ground 7, Petitioner asserts that claims 15 and 17–19 would have been obvious over Native, Bianchi ’254, Easy Homemade, Anconi, and Stoia. Pet. 54–60. For Ground 8, Petitioner asserts that claim 16 would have been obvious over Native, Bianchi ’254, Easy Homemade, Anconi, Stoia, and Millet. Pet. 60–63. To support its contentions, Petitioner provides detailed claims charts and/or discussion for each of these grounds explaining how each claim limitation is disclosed in the asserted references and how the claims recite deodorants formulated by combining well-known deodorant ingredients with obvious design choices. *See* Pet. 41–50 (Ground 5), 53–54 (Ground 6), 57–60 (Ground 7), 63 (Ground 8). Petitioner further supports its assertions with the testimony of Dr. Michniak-Kohn. Ex. 1002 ¶¶ 148–244.

After considering all of the arguments and cited evidence of record, we determine that Petitioner has shown that the preponderance of the evidence supports the conclusion that claims 1–19 would have been obvious over Lesniak in combination with Bianchi ’254, Phinney, Anconi, and Easy Homemade. Our reasoning follows.

1. *Ground 5: Obviousness over Native and Bianchi ’254 (Claims 1, 4–14)*

a) *Petitioner’s Contentions*

Petitioner contends that Native

> discloses features of deodorant stick compositions claimed in the ’752 Patent; however, it does not expressly teach the percentage of each ingredient used, including the percentage of liquid triglyceride (carrier oil). Native also does not disclose the hardness of the deodorant stick. Bianchi ’254 describes antiperspirant stick compositions, which establish that the claimed concentrations of ingredients were conventional and that the claimed hardness range was standard. Accordingly, it would

> be obvious to formulate a deodorant stick such as the stick disclosed by Native using ingredient concentrations and hardnesses as taught by Bianchi '254.

Pet. 39 (citing Ex. 1002 ¶¶ 148–49).

b) Patent Owner's Contentions

Patent Owner provides four reasons why the combination of Native and Bianchi '254 fails. First, Patent Owner asserts that Native's deodorant stick was "a known leader in the field," and Native discloses only positive reviews for Native's deodorant stick. PO Resp. 24. Consequently, Patent Owner asserts, "[w]ith this level of consumer acceptance and praise, and no identified issues or concerns, there was simply no reason for a POSA to have sought to modify Native." *Id*. at 52 (citing Ex.2011 ¶ 119). Any such modification, Patent Owner asserts, is impermissibly based on hindsight using the '752 patent as a roadmap. *Id*. at 52, 60.

Second, Patent Owner points to the differences in the Native's aluminum-free product containing natural ingredients and Bianchi '254's formulation containing volatile silicon and aluminum. PO Resp. 53. Patent Owner concludes that a "POSA would not have been motivated to consider the teachings of Bianchi-254 in combination with Native, as Bianchi-254's prominent use of non-natural ingredients and aluminum would defeat the core objective of Native to provide aluminum-free, natural deodorant products." *Id*. (citing Ex. 2011 ¶ 118).

Third, Patent Owner contends that, "[j]ust like the Lesniak-based grounds, Petitioner fails to show its Native combinations disclose the hardness range required by all claims." *Id.* Similar to its arguments rebutting the Lesniak-based grounds, Patent Owner argues that "Bianchi-254's hardness measurements result from use of a 100-g total load needle assembly, which is twice the total load used by the ’752 Patent’s needle assembly.” *Id.* at 54.

Fourth, Patent Owner again points to the differences between the Native and Bianchi ’254 formulations concluding:

> The Petition does not explain what would have motivated a POSA to have implemented the weight-percentages from a non-natural antiperspirant stick into the natural, aluminum-free stick of Native (if it is even possible), or what a POSA would have substituted for aluminum and volatile silicones in order to preserve the same weight percentages without the presence of the aluminum active and silicones.

PO Resp. 54 (citing Ex. 2011 ¶¶ 119).

c) Petitioner Reply

Petitioner responds that it is not necessary that Native itself identify any deficiencies for a POSA to be motivated to modify it, and also points to Easy Homemade as providing explicit motivation to modify Native. *See* Pet. Reply 18 (citing *KSR Int’l*, 550 U.S. 398, 418–21). Petitioner also responds that the differences between the formulations of Native and Bianchi ’254 would not prevent a POSA from combining the teachings because

> Native already achieved a silicone and aluminum-free deodorant stick, so a POSA therefore would not have to draw on Bianchi-254’s teachings for those features. Ex. 1002, 69-72. Native further discloses the relative amounts of each ingredient in that stick. *Id*. Bianchi-254 simply provides a POSA with guidance on the quantitative amounts of those ingredients and preferred hardness.

Pet. 20. Petitioner points out that the Specification of the ’752 patent in fact “incorporates references disclosing aluminum-containing antiperspirants for their disclosure of ‘suitable’ structurants, emulsifiers, antimicrobials, and other ingredients." *Id.* at 21 (citing Ex. 1001, 16:16–19, 19:19–26; Ex. 1041; Ex. 1083; Ex. 1081, 46:22–52.22).

d) Patent Owner Sur-reply

Patent Owner reiterates that the differences in formulations between Native and Bianchi '254 indicate a POSA would need to use impermissible hindsight to arrive at the claimed inventions. PO Sur-Reply 23–24. Patent Owner explains that "Petitioner ignores that Bianchi-254's 'guidance' on the quantitative amounts of the ingredients in question is influenced and scaled by the presence of over 50% aluminum and silicone in Bianchi's sticks." *Id.*

e) Discussion

Petitioner asserts that Native's natural, aluminum-free deodorant stick is comprised of the following ingredients—caprylic/capric triglyceride, baking soda, stearyl alcohol, hydrogenated castor oil, polyglycerol-3 beeswax, and jojoba esters—although the percentage of each ingredient is not disclosed. *See* Pet. 41–50 (citing Ex. 1007, 5, 11).[4] Dr. Michniak-Kohn testifies that "[i]n the absence of express teachings on these points, a POSA would have looked to other publications in the field with suitable examples." Ex. 1002 ¶ 150.

Dr. Michniak-Kohn notes that Native's primary structurant is a fatty alcohol as in Bianchi '254's antiperspirant stick compositions. Ex. 1002 ¶ 151. Dr. Michniak-Kohn also notes that "Bianchi '254 singles out stearyl alcohol as a 'widely used' structurant 'of particular interest' for deodorant." *Id.* ¶¶ 72, 169 (citing Ex. 1008 ¶¶ 1, 58). Dr. Michniak-Kohn concludes that "[w]hen formulating a natural, aluminum-free deodorant stick based on the

[4] Dr. Michniak-Kohn notes that aluminum salts and synthetic fragrances are not included in Native's composition. *See* Ex. 1002 ¶ 150.

composition in Native, a POSA would have looked at publications describing other stick compositions, like Bianchi ’254, for information regarding the amount of ingredients.” *Id.* ¶ 152. We credit Dr. Michniak-Kohn’s testimony as consistent with the record evidence, and further find unpersuasive Dr. Lochhead’s testimony to the contrary.

First, we note that Petitioner looks to the teachings of Bianchi ’254 to inform the appropriate amount for the ingredients described in Native, not to alter Native’s formulation. *See* Pet. 39 (stating “it would be obvious to formulate a deodorant stick such as the stick disclosed by Native using ingredient concentrations as taught by Bianchi ’254”). Therefore, Patent Owner’s argument that the praise for the Native product would not incentivize a POSA to change the Native formulation misses the mark. Petitioner does not seek to change Native’s formulation, but verify the amount of the listed ingredients. *Id.* In any event, as Petitioner notes, a reason to arrive at a claimed invention does not have to come from a particular identification of deficiencies in Native itself; as the Supreme Court noted in *KSR* such motivation may come from “any need or problem known in the field of endeavor at the time of invention and addressed by the patent.” 550 U.S. at 420; Pet. Reply 18.

As Petitioner notes, both Native and Bianchi ’254 are in the same field as the ’752 patent—personal care compositions, including deodorants and/or antiperspirants. Pet. 43 (citing Ex. 1001, Abstract, 1:10–2:57, 9:65–10:4; Ex. 1007, 5–8, 11; Ex. 1008, Abstract, ¶¶ 10–21; Ex. 1002 ¶ 155). There is nothing in the record that we have found, nor has Patent Owner pointed us to any such evidence, to distinguish natural, aluminum-free deodorants from antiperspirants containing aluminum in such a way that ingredients of one would not be used for the same purpose in the other. As the Supreme Court noted in *KSR*, "[t]he combination of familiar elements according to known methods is likely to be obvious when it does no more than yield predictable results." 550 U.S. at 416. In fact, as Petitioner points out, the '752 patent "incorporates references disclosing aluminum-containing antiperspirants for their disclosure of 'suitable' structurants, emulsifiers, antimicrobials, and other ingredients." Pet. at 21 (citing Ex. 1001, 16:16–19, 19:19–26; Ex. 1041; Ex. 1083; Ex. 1081, 46:22–52:22).

Dr. Michniak-Kohn persuasively asserts on the record before us that when looking at particular ingredients of a formulation, a POSA would look to other references in the same field—personal care compositions, including deodorants and/or antiperspirants—to determine how much of each ingredient should be used. We are not persuaded by the testimony of Dr. Lochhead to the contrary. Dr. Lochhead testifies that:

> In my opinion, a POSA seeking to develop an aluminum-free and silicone-free deodorant stick would not have been motivated to implement the weight percentages disclosed in Bianchi-254, which describes and claims antiperspirant formulations containing about 50% or more by weight of aluminum and volatile silicones. The Petition does not mention Bianchi-254's inclusion of these ingredients, despite its sole reason for combining Native and Bianchi-254 hinging on it: "In making a natural, aluminum-free deodorant stick composition, a POSA would have looked to existing compositions, such as Native" and would look at other "suitable examples" such as Bianchi-254, which "would have been of particular use for a POSA looking for percentages to use for Native's formulation." (Petition at 39-40.)
>
> Bianchi-254 does not disclose a "natural, aluminum-free deodorant stick." The Petition does not explain what would have motivated a POSA to have implemented the weight percentages

> from a non-natural antiperspirant stick into the natural, aluminum-free stick of Native, or what a POSA would have substituted for aluminum and volatile silicone in order to preserve the same weight percentages without the presence of the aluminum active and silicones. The Petition also does not explain how a POSA would have had a reasonable expectation of success in implementing Bianchi-254's teachings while maintaining a stable composition.

Ex. 2011 ¶¶ 128–129.

Dr. Michniak-Kohn provides a convincing rationale for a POSA to look to Bianchi-254 for weight-percentages for the ingredients of Native even though Bianchi-254 is not a "natural" deodorant. Dr. Michniak-Kohn notes that Native's primary structurant is stearyl alcohol, a fatty alcohol, and Bianchi '254 "is a publication in the field which describes 'fatty-alcohol containing antiperspirant stick compositions' and methods related thereto." Ex. 1002 ¶ 151 (citing Ex. 1008, Abstract, ¶¶ 10–12, 21).

Dr. Michniak-Kohn further persuasively testifies that the combination of Native and Bianchi '254 teaches the "at least 25% (or 40%) by weight liquid triglyceride" limitations. Dr. Michniak-Kohn states:

> Bianchi '254 describes an antiperspirant stick with 40-80% oil. In Bianchi '254, part of the oil carrier (20-40% of the total) is a natural oil such as a liquid triglyceride. Ex. 1008, ¶¶ [0017], [0050]. Bianchi '254 also describes that the antiperspirant stick further comprised 20-40% of silicone oil. *Id.*, ¶ [0016]. A POSA would have understood that silicone oil and liquid triglyceride both functioned as "carrier oils" that, among other things, solubilize and disperse other ingredients (id., ¶¶ [0015], [0023], cl. 17, 18), and therefore would have been motivated to use even more liquid triglyceride when formulating a natural deodorant stick (like in Native) that did not include, e.g., silicone oil, to meet what the '752 Patent admits was an existing consumer preference (Ex. 1001, 1:10-24).

> Thus, in my opinion, a POSA would have been motivated to use from about 25% to 50% of a liquid triglyceride in a deodorant stick, and would have had a reasonable expectation of success in doing so, inasmuch as (i) Bianchi '254 already suggests the same with its 20-40% amount, (ii) a POSA would have, based on Bianchi '254's overall oil teachings, used even more of the liquid triglyceride when making a stick with only that oil carrier (as taught by Native).

Ex. 1002 ¶¶ 178–179. We agree with this assessment of the combination of the teachings of Bianchi '254 and Native. A POSA would have increased the amount of liquid triglyceride in Bianchi '254 in the 40 to 80% range for the total amount of oil disclosed in Bianchi '254 when applied to the Native formulation that is natural and does not contain any silicone oil to arrive at the at least 25% to 40% by weight liquid triglyceride.

We also agree that this conclusion is supported by the overlapping ranges for oil disclosed in Bianchi '254 and the challenged claims. *See* Ex. 1002 ¶ 180. Further, we agree with Dr. Michniak-Kohn that

> there was nothing critical or unexpected about the claimed range relative to Bianchi '254's ranges. As a POSA would have understood, the minor deviations in the ranges yield predictable variations in degree and do not produce a different kind of result. For example, using more liquid triglyceride generally yields a softer stick, merely changing the degree of hardness.

Id. ¶ 180. Patent Owner offered no rebuttal to this assessment that changing the amount of liquid triglyceride is merely a variation in degree and not result. *See KSR*, 550 U.S. at 416 (stating "combination of familiar elements according to known methods is likely to be obvious when it does no more than yield predictable results").

We determine that Petitioner has shown by a preponderance of the evidence that claims 1–13 would have been obvious over Native and Bianchi ’254.

2. *Ground 6: Obviousness over Native, Bianchi ’254, and Easy Homemade (Claims 2–3)*

Petitioner asserts that claims 2 and 3 would have been obvious over Native in view of Bianchi ’254 and Easy Homemade. Pet. 50–54. Petitioner relies on the explanation of how the limitation of claim 1 are met by the teachings of Native and Bianchi ’254, but notes that Native does not teach "a deodorant stick composition (1) substantially free of baking soda (Native's antimicrobial); or (2) comprising antimicrobials with a water solubility of at most about 50 or 90 g/L at 25°C." Pet. 50. Petitioner asserts that Native's antimicrobial, baking soda, was known to cause skin irritation for some users, and Easy Homemade explains that magnesium hydroxide with a water solubility within the claimed range is an antimicrobial that works just as well as baking soda. Pet. 50–51 (citing Ex. 1009, 6, 9).

Many of the issues raised by Patent Owner in response to Ground 6 were also raised with regard to Grounds 4 and 5. For example, as in Ground 4, Patent Owner challenges the public accessibility of Easy Homemade and whether Easy Homemade teaches the removal and substitution of baking soda. *See* PO Resp. 60. We rely on our analysis set forth in Grounds 4 and 5 and provide the following additional analysis.

Patent Owner asserts that Petitioner misreads Easy Homemade as teaching the removal and substitution of baking soda, but,

> Rather, the Easy Homemade hobbyist tells her followers to purchase a pre-made baking soda-free deodorant base—a base Petitioner's expert could not identify the ingredients of—and add

> some magnesium hydroxide powder to it. As Petitioner's expert confirms, Easy Homemade "doesn't talk about removing any ingredients from these sticks; she's talking about adding ingredients."

PO Resp. 60 (citing Ex. 1009, 7, Ex. 1012, 120:6–8, 121:1–9; Ex. 2011 ¶ 132).

Petitioner responds that using prior art elements according to their established functions is obvious. Pet. Reply 21–22 (citing *KSR*, 550 U.S. at 417; *Intel Corp. v. PACT XPP Schweiz AG*, 61 F.4th 1373, 1380-81 (Fed. Cir. 2023)). Therefore, Petitioner asserts that

> A known antimicrobial for deodorants, baking soda, caused irritation for some users, and another known antimicrobial for deodorants, magnesium hydroxide, ameliorated that issue when substituted for baking soda. 'Nothing more is required to show a motivation to combine under *KSR*.'"

Id. at 22 (citing *Intel*, 61 F.4th at 1381). We agree.

Dr. Michniak-Kohn points to evidence that a POSA would have known that deodorants containing baking soda could cause skin irritation for some users and Easy Homemade confirms what POSA would have known—namely, that magnesium hydroxide was a known substitute for baking soda that avoided skin irritation. Ex. 1002 ¶ 186 (citing Ex. 1009, 6, 9). Petitioner further notes that Easy Homemade expressly discusses substituting magnesium hydroxide for baking soda because Easy Homemade states that "baking soda can easily be swapped for magnesium hydroxide." Reply 22 (citing Ex. 1009, 6, 9). We agree with Petitioner that "Easy Homemade just directs the POSA to substitute one known antimicrobial for another." *Id.*

Based on the full record before us, we determine that Petitioner has shown by a preponderance of the evidence that claims 2 and 3 would have been obvious over Native, Bianchi ’254, and Easy Homemade.

3. *Ground 7: Obviousness over Native, Bianchi ’254, Easy Homemade, Anconi, and Stoia (Claims 15, 17–19)*

Petitioner asserts that claims 15 and 17–19 would have been obvious over Native, Bianchi ’254, Easy Homemade, Anconi, and Stoia. Pet. 54–60. Petitioner states:

> While Native teaches a deodorant stick composition comprising an antimicrobial and various plant compounds, it does not teach a deodorant stick comprising (1) triethyl citrate; or (2) sunflower seed oil. But a POSA would have found it obvious to use these compounds in Native’s deodorant based on Anconi and Stoia.

Pet. 54–55 (citing Ex. 1002 ¶ 196).

Petitioner contends that Anconi teaches triethyl citrate as having antibacterial and deodorant properties, and that a POSA would have reason to use it instead of baking soda that may cause skin irritation with a reasonable expectation of success. Pet. 55 (citing Ex. 1057 ¶ 27). Petitioner cites Stoia for teaching tocopherols that have antioxidant properties are abundant in sunflower seeds and that sunflower seed oil may be used as an emollient that a POSA would have been motivated to use for its “positive skin effects” with a reasonable expectation of success. Pet. 56 (citing Ex. 1058, 45; Ex. 1002 ¶¶ 197–200).

Similar to arguments raised in the Lesniak grounds, Patent Owner emphasizes the differences in the composition of Native and Anconi (Native is an anhydrous, wax-structured composition with a triglyceride-based carrier system, like Lesniak, but Anconi is an aqueous propylene glycol- and/or alcohol-based mixture solidified by sodium sterate) to support its argument that a POSA would not have reason to reformulate Native with teachings from a "different stick technology." PO Resp. 63.

We rely on our previous analysis for these issues, and determine that a POSA would have had reason to use triethyl citrate in lieu of or addition to the baking soda of Native, and also would have had reason to add sunflower seed oil for its antioxidant properties and positive skin effects. We find that these additions to the Native formulation to be merely a combination of familiar elements according to known methods that yield predictable results, and therefore, would have been obvious. *See KSR*, 550 U.S. at 416.

Based on the full record before us, we determine that Petitioner has shown by a preponderance of the evidence that claims 15 and 17–19 would have been obvious over Native, Bianchi '254, Easy Homemade, Anconi, and Stoia.

4. Ground 8: Obviousness over Native, Bianchi '254, Easy Homemade, Anconi, Stoia, Millet (Claim 16)

Petitioner asserts that claim 16 is unpatentable under 35 U.S.C. § 103 as obvious over the combination of Native, Bianchi '254, Easy Homemade, Anconi, Stoia, and Millet. Claim 16 depends from claim 15 and further requires that the recited deodorant stick of claim 15 also include tapioca starch. Petitioner contends that Millet teaches starch as an "effective filler in deodorants capable of absorbing sweat and moisture and that tapioca starch is specifically preferable." Pet. 60–61 (citing Ex. 1044, ¶¶ 38, 39, 42). Accordingly, Petitioner alleges it would have been obvious to use tapioca starch in the Native deodorant stick, as taught by Millet. *Id.* (citing Ex. 1002 ¶ 238).

Patent Owner contends that "Petitioner has not identified any problem with Native that would have motivated a POSA to reformulate it" by substituting tapioca starch for Native's arrowroot powder. PO Resp. 66–67. Patent Owner further contends that Petitioner's reason for adding starch falls short because Native teaches the use of arrowroot powder, which serves the stated function of absorbing moisture and sweat. *Id.* (citing Ex. 2011 ¶ 147). Patent Owner also points to alleged instability issues as we addressed for the Lesniak grounds.

Having considered the parties' positions and evidence of record, summarized above, we agree with Petitioner's position, which we adopt as our own. Pet. 60–63; Ex. 1002 ¶¶ 238–44; *see also Almirall*, 28 F.4th at 273 (obvious to "substitute one known" option for another). Accordingly, we determine that Petitioner has shown by a preponderance of the evidence that claim 16 is rendered obvious by the combination of Native, Bianchi '254, Easy Homemade, Anconi, Stoia, and Millet.

E. Squatch/Sturgis Ground (Ground 9 – Claims 15–19)

Petitioner asserts that claims 15–19 of the '752 Patent would have been obvious over Squatch and Sturgis. Pet. 67–72. Petitioner asserts that Sturgis is prior art because "claims 15–19 are not entitled to a priority date before the November 3, 2022, filing date" of the '752 patent. Pet. 63. Petitioner also asserts that Squatch is a printed publication because it "is a website excerpt disclosing known aluminum-free deodorant formulations that comprise caprylic-capric triglyceride, arrowroot powder, stearyl alcohol, and magnesium hydroxide," that "published on a well-known website, which received over 12.5 million views in November 2020." *Id.*

We agree with Patent Owner that Petitioner has not shown sufficiently that Squatch is a printed publication. As described above, Petitioner relies on Squatch as a "website excerpt" placed on its site at the relevant time, but has not produced a screen shot of the website to show that it ever existed. *See* Ex. 1059. Petitioner provides internal documents that purport to describe what would have been on Dr. Squatch's website at the relevant time, but falls short of establishing what was on the website for the interested consumer to view. Thus, we determine that this ground fails.

III. CONCLUSION[5]

Based on the fully developed trial record, Petitioner has demonstrated by a preponderance of the evidence that claims 1–19 of the '752 patent are unpatentable.

[5] Should Patent Owner wish to pursue amendment of the challenged claims in a reissue or reexamination proceeding subsequent to the issuance of this decision, we draw Patent Owner's attention to the April 2019 *Notice Regarding Options for Amendments by Patent Owner Through Reissue or Reexamination During a Pending AIA Trial Proceeding*. *See* 84 Fed. Reg. 16,654 (Apr. 22, 2019). If Patent Owner chooses to file a reissue application or a request for reexamination of the challenged patent, we remind Patent Owner of its continuing obligation to notify the Board of any such related matters in updated mandatory notices. *See* 37 C.F.R. § 42.8(a)(3), (b)(2).

In summary:

Claim(s)	35 U.S.C. §	Reference(s)/ Basis	Claim(s) Shown Unpatentable	Claim(s) Not Shown Unpatentable
1–13	103	Lesniak, Bianchi ’254	1–13	
14	103	Lesniak, Bianchi ’254, Phinney	14	
15,16, 18, 19	103	Lesniak, Bianchi ’254, Phinney, Anconi	15, 16, 18, 19	
17	103	Lesniak, Bianchi ’254, Phinney, Anconi, Easy Homemade	17	
1, 4–14	103	Native, Bianchi ’254	1, 4–14	
2–3	103	Native, Bianchi ’254, Easy Homemade	2–3	
15, 17–19	103	Native, Bianchi ’254, Easy Homemade, Anconi, Stoia	15, 17–19	
16	103	Native, Bianchi ’254, Easy Homemade, Anconi, Stoia, Millet	16	
15–19	103	Squatch and Sturgis		15–19
Overall Outcome			1–19	

IV. ORDER

In consideration of the foregoing, it is

ORDERED that claims 1–19 of the ’752 patent have been shown to be unpatentable; and

FURTHER ORDERED that, because this is a Final Written Decision, parties to this proceeding seeking judicial review of our decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

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